

Small-Scale Wetland Restoration in the High Arctic: A Long-Term Perspective

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Abstract

Results are presented using vegetative shoots and bryophyte sods to restore floristically impoverished high arctic wet sedge-moss meadows that had suffered intense damage from vehicle activity during the period 1960–1967. Clonal transplants of *Carex aquatilis* var. *stans*, a native sedge, were planted with and without bryophyte sods in vehicle ruts in 1972. After nearly two decades, there was less *Carex* cover in the planted ruts with flowing water than in the contiguous controls. This pattern was slightly reversed in planted plots with standing water. Reinvasion of *Eriophorum angustifolium* occurred in treated ruts, but cover was less in both treatments than in controls in 1990. The unexpected recruitment of *Eriophorum scheuchzeri* from the seed bank in moss-sodded plots is discussed in terms of its local and regional importance. Total plant cover in restored ruts was nearly equal to that of controls, but biomass was somewhat less than that in control plots. Plots with bryophytes were environmentally distinct, due primarily to increases in organic mat depth relative to controls. After 18 years, restoration efforts resulted in increased plant cover in treated ruts compared to naturally recovering ruts.

The composition of no two patches of vegetation is precisely the same [and] neither are the seed banks. Succession on different patches of disturbed ground in the same locality frequently proceeds quite differently because of such differences.

—J. Miles, *Vegetation Dynamics*, 1979

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According to Gross (1987), "one of the distinctive challenges of ecological restoration, as opposed to less ambitious forms of land reclamation, is the creation of communities with the relatively high level of species diversity that typifies many natural communities." As Bradshaw (1987) points out in the same volume, the acid test of our understanding of the factors that determine diversity in natural communities will be our ability to restore or rebuild a reasonable facsimile of a community that has been damaged or destroyed (Gross 1987). Tundra communities are considered relatively simple in structure and depauperate of vascular floristics, particularly in more climatically and edaphically severe high arctic zones (Young 1971; Edlund 1990), although regional cryptogamic species richness tends to increase in these zones relative to temperate latitudes (Longton 1988). Wetlands are extremely limited in the High Arctic (Babb & Bliss 1974) and are especially poor in terms of both vascular and nonvascular floristics. Since these ecosystems are low in species diversity, they theoretically represent a somewhat smaller challenge for restoration ecology. This view may be overly simplistic, however, in a region with such unique hydrological features (Woo & Steer 1986).

Cairns (1986) observed that small disturbances caused by natural phenomena are often followed by a rapid recovery process, although disturbances that are entirely the result of human activity are often not so quickly healed. The Arctic, in general, has been noted for its slow rates of recovery from disturbance (Hanson & Lamson 1990), and recovery in the High Arctic, in particular, is presumed to be slower than in the Low Arctic (Babb & Bliss 1974). The major restoration efforts in Alaska have been considered by some to be largely successful (Alexander & Van Cleve 1983; Walker et al. 1987a), although serious cumulative effects are now evident in the most extensively disturbed areas (Walker et al. 1987b). A recent investigation in the High Arctic provides strong evidence questioning whether recovery from vehicle impact in wetlands occurs on a human time scale under natural circumstances (Forbes 1992a), while acknowledging that recovery is "a somewhat loaded word" (McIntosh 1980). The aim of this paper is to provide long-term observations of arctic wetland restoration after local disturbance from tracked vehicles. Such observations are important in a primarily vegetative region where plant growth is slow, and short-term observations may not be indicative of long-term trends.

Study Site. The study site is Truelove Lowland (43 km²) (75°33'N, 84°40'W) on the north coast of Devon Island (54,000 km²), Canada, well known from its use as an intensive study site during the International Bio-



Figure 1. General aspect (looking east) of the wet sedge/moss meadow on Truelove Lowland, showing a more recent set of tracks with standing water. In low-lying areas such as this, the substrates are mostly dolomite-derived, so that minerotrophic fens develop wherever there is an adequate water supply. Granitic outcrops can be seen beyond the pond and the dolomite plateau in the far background. Note the lack of hummocky microtopography in the meadow. Experimental and control plots are immediately to the rear of the photographer. Photo taken July 27, 1990.

logical Programme (IBP) (Bliss 1977). In the gently sloping wet meadows where this study occurred (Fig. 1), a single species, *Carex aquatilis* var. *stans* (arctic water sedge) was dominant, with *Eriophorum angustifolium* (common cotton-grass) an important associate. These two plants appear morphologically similar but occupy different niches due to divergent rooting strategies and preferences for different peat depths (Daly et al. 1988; Callaghan et al. 1991). The grasses *Arctagrostis latifolia* ssp. *latifolia* (polar grass), *Dupontia fisheri* ssp. *psilosantha*, *Juncus biglumis* (two-flowered rush), *Saxifraga hirculus* (yellow marsh saxifrage), and *Pedicularis sudetica* (lousewort) are also frequent but contribute relatively little in terms of cover or biomass (Muc 1977). *Eriophorum scheuchzeri* (arctic cotton-grass) is common within the investigated meadows, but mature flowering individuals are rare (Barrett 1975).

Among associated bryophytes in these same meadows, 78% of the moss cover is provided by only three species (*Calliergon giganteum*, *Drepanocladus revolvens*, *Meesia triquetra*), taxa indicative of the prevalent minerotrophic conditions (Vitt & Pakarinen 1977). Mosses can provide up to 75% of the ground cover in some of the wettest meadows where live vascular cover is minimal (approximately 25%). These lowland meadows have been floristically classified by Barrett (1972) into one phytosociological alliance, *Caricion aquatilis*, which is further subdivided into one association, *Caricetum stantis*, and two subassociations, *caricetosum stantis* and *caricetosum membranacei*. These subassociations correspond roughly to Muc's (1977) physiognomic classes of wet sedge-moss and hummocky meadows, respectively. The sampled meadows

discussed here are entirely within the wetter *caricetosum stantis* subassociation (Fig. 1), differentiated by the absence of *Carex membranacea*. Lichens are regionally abundant but contribute less than 1% plant cover in these meadows. These wet lowland landscapes cover less than 1% of the High Arctic and are considered critical habitat for the region's terrestrial herbivores (Babb & Bliss 1974). The Truelove Lowland is moderately grazed by muskoxen (Henry 1992).

Methods

The investigated disturbances from tracked vehicles date from the 1960s and were originally documented by Barrett (1975). In July 1972, Barrett established 19 treatment blocks (0.5×5 m) in which vegetative shoots of *Carex aquatilis* var. *stans* were transplanted from undisturbed sites into adjacent, severely rutted, saturated meadows. Most of the results reported here are from two of these experimental blocks, redefined here as plots: one rut on mineral soil known to have flowing water during spring melt at the time of planting where 55 tillers were planted (Treatment 1, Fig. 2), and another similar rut in an area of standing water where 45 tillers were planted (Treatment 2, Fig. 3). In Treatment 1, the gaps among the vegetative shoots of *Carex* were filled in with sods of bryophytes from the adjacent meadow so that the rut was brought to the level of the surrounding undisturbed wetland community. Barrett's aim was to determine if the cloned transplants of native species would survive and grow in denuded areas and thus serve as an alternative to



Figure 2. Saturated experimental plots on Truelove Lowland in July 1972 showing the pattern of planting for vegetative shoots of *Carex aquatilis* var. *stans* interspersed with sods of bryophytes (Treatment 1). The tracks have gently running water during spring runoff. It was the use of such sods that apparently led to the recruitment of *Eriophorum scheuchzeri* from the seed bank, as shown in Figure 5. Photo by P. E. Barrett.

seeding, since so few species produce viable seed in extreme tundra environments (Bliss & Grulke 1988). He revisited the sites in August 1973 and determined that 60% of the *Carex* plants had survived in the rut with flowing water and that 13% of these showed new growth. In the standing water rut (Treatment 2), 69%

of the plants had survived and 36% exhibited new growth.

I resurveyed the study areas in July and August of 1990. In addition to transplanted sites, several similarly disturbed but naturally recovering wet and mesic meadows, some immediately adjacent to the at-

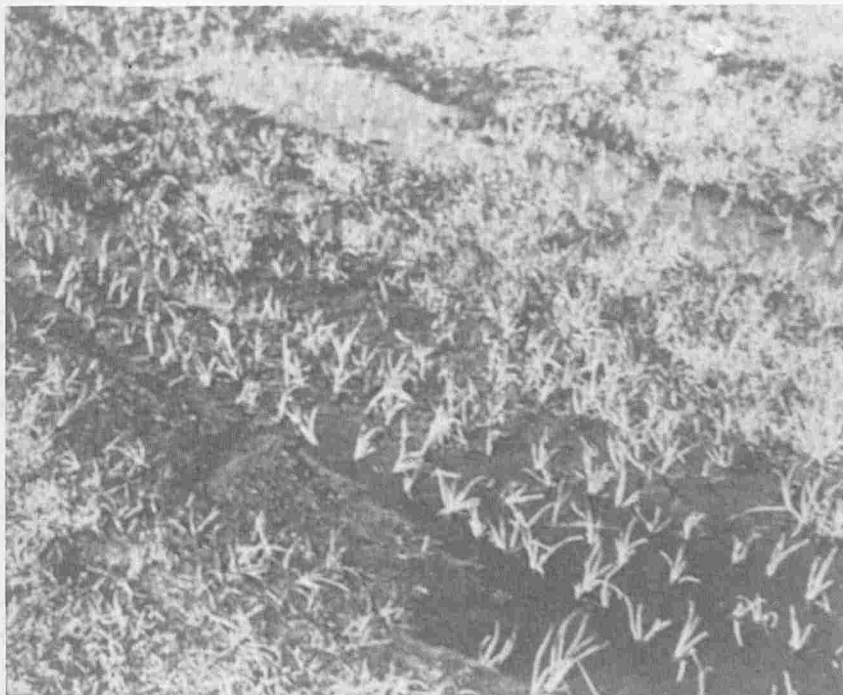


Figure 3. In Treatment 2, only vegetative shoots of *C. aquatilis* var. *stans* were transplanted into ruts with standing water and no bryophytes were added. Photo taken in July 1972 by P. E. Barrett.

tempted restorations, were also sampled, as well as undisturbed wetland plots as controls (Fig. 1, Table 1). Some naturally recovering plots were sampled as far away as the Clyde River on NE Baffin Island (Forbes 1992a, 1992b). These stands were included in the analysis and discussion for comparative purposes, as recommended by Cairns (1987). Stratified random sampling was employed using 25 × 50 cm quadrats. Each study plot was comprised of five quadrats. These were situated in (1) restored ruts, (2) in naturally recovering ruts, and (3) in three immediately adjacent control areas (within 4 m) considered to be relatively undisturbed. Percent cover and biomass of the vascular plants were measured, as well as the composition and frequency of bryophytes. Aboveground live and dead vascular standing crop was measured at the peak of the growing season. Other biotic measurements included culm height, counts of graminoid inflorescences, and depth of organic mat. Nomenclature for vascular plants follows Porsild and Cody (1980) and for bryophytes follows Vitt (1975).

The pH of mineral, and occasionally highly organic, soils and surface waters was measured in the field for each quadrat using a Cole Parmer model 5985-80 DigiSense™ meter accurate to ±0.05 according to methods given in Sheldrick (1984). Cores of 100 cm³ were taken from fresh mineral soils at a depth of 5–10 cm and oven dried at 105°C for 24 hours in the field. These were analyzed for conductivity (corrected for H⁺ and temperature) at Macdonald College, Ste. Anne de Bellevue, Quebec (Richards 1969).

Data Analysis. Both biotic and abiotic variables were tested for normality by using moments reports and standardized skewness and kurtosis measures (Zar 1984; SAS Institute, Inc. 1989). Raw cover/abundance and biomass values in the wet meadows were normally distributed means and were compared using the parametric *t*-test (Zar 1984; BrainPower, Inc. 1985). Log-transformed vegetation cover/abundance data were classified using a hierarchical polythetic divisive technique. The species data were also ordinated using detrended correspondence analysis, after which vegetation-environment relationships were analyzed using canonical correspondence analysis. The multivariate analyses were of limited utility given the great floristic and environmental similarity among the plots; the rationale behind the programs is therefore not discussed here in detail.

Results

The wet meadows on Truelove Lowland had a pH range in undisturbed mineral soils from 6.1 to 7.7, with

a range from 6.6 to 8.1 in surface waters. Corrected conductivities in undisturbed mineral soils ranged from 140 to 245 μ S. These levels were substantially higher than and often more than twice the levels measured in transitional poor fens (Sjörs 1952) on Baffin Island, indicating minerotrophic fen conditions in and around the experimental plots. Thus, the rich fens are characterized primarily by the bryophytes mentioned earlier, in addition to *Campylium arcticum*, *Catocarpium nigrum*, *Cinclidium arcticum*, and *Drepanocladus brevifolius*, all of which have moderate or strong fidelity to calcareous substrates and typically reach their highest frequencies in wet sedge meadows (Barrett 1972; Vitt 1975). Evidence of moderate grazing by muskoxen included random clumps of scat and tufts of "qiviut," or long, silky hairs, that clung to quadrat-marking stakes the oxen had brushed against. Grazing within the plots appeared indiscriminate, with the exception of the few flowering *Pedicularis* plants, which were selectively clipped just after anthesis.

When Barrett (1975) resurveyed the plots in 1973, he did not detect *E. scheuchzeri*. However, it is possible that viable seeds transported in the bryophyte sods had already germinated but were not readily recognized, because immature plants are easily confused with *E. angustifolium* (Barrett 1975). The highly local and striking abundance of *E. scheuchzeri* inflorescences stood out conspicuously against the dull green monochrome of the flat wetland.

In comparing the restored plot with gently flowing water and bryophytes added (Treatment 1) to the adjacent control, there was 7.0% less cover of *Carex* (Table 1) in the restored rut, which was small but consistent and apparently significant compared to the control ($p = 0.025$, *t*-test, $n = 5$ per treatment and control). There was also a significant reduction (4.0%; $p = 0.025$) in cover of *E. angustifolium*, but a highly significant increase ($p = 0.0005$) of *E. scheuchzeri*, which was completely absent in all the control plots and other meadows on the lowland (Table 1). The minute *Ranunculus hyperboreus* (buttercup) occurred at low cover values in both treatment and control areas but with an increase ($p = 0.025$) in frequency on the experimental plot with flowing water (Table 1).

In Treatment 2, with standing water and the absence of moss sods, *Carex* cover was about the same as the contiguous control (18% versus 16%; Table 1). However, cover of *E. angustifolium* was threefold less than the control (2% versus 6%), and *E. scheuchzeri* was absent. Total plant cover in Treatment 2 was only about half that in Treatment 1, with flowing water and bryophytes added (20% versus 43%). Total plant cover in Treatment 1 was nearly equal to that of the control (43% versus 47%), whereas in the naturally recovering rut cover it was less than the control (17% versus 24%).

Table 1. Mean percent cover in restored (R) and naturally recovering ruts (NR) after 19 years, along with adjacent controls (C).

Species	Study Plots					
	1		2		3	
	R	C	R	C	NR	C
<i>Carex aquatilis</i> var. <i>stans</i>	21.4	28.6	18.0	14.4	16.2	17.4
<i>Eriophorum angustifolium</i>	10.0	14.2	1.8	5.8	0.1	3.8
<i>Ranunculus hyperboreus</i>	0.6	0.1	0.2	0.1	—	0.1
<i>Arctagrostis latifolia</i> ssp. <i>latifolia</i>	—	0.6	—	0.2	0.2	0.8
<i>Juncus biglumis</i>	3.4	2.4	—	0.2	—	0.2
<i>Saxifraga hirculus</i>	0.2	0.6	—	0.2	—	0.2
<i>Dupontia fisheri</i> ssp. <i>psilosantha</i>	—	—	—	0.3	0.2	1.0
<i>Eriophorum scheuchzeri</i>	7.6	—	—	—	—	—
<i>Salix arctica</i>	—	—	—	—	—	0.4
<i>Pedicularis sudetica</i>	—	—	0.2	—	—	—
Total Plant Cover	43.2	46.7	20.0	21.3	16.7	23.9
Dead Shoots/Leaves	4.8	14.8	5.4	13.2	6.0	7.8

1 = Treatment 1, with seasonally flowing water in which both cuttings and bryophytes were planted versus control; 2 = Treatment 2, with standing water in which only cuttings (no bryophytes) were planted versus control; 3 = Plot with standing water—naturally recovering ruts versus control.

This meant that the naturally recovering tracks were still clearly visible.

Apart from *E. scheuchzeri*, graminoids exhibited no significant differences in culm heights between disturbed and control plots. Inflorescence density counts, however, revealed a significantly higher number of flowering culms of *Carex* in Treatment 1 than in the undisturbed control. The density of flowering culms of *E. scheuchzeri* in the same plot was also highly significant ($p = 0.0005$) relative to the control.

Analyses of aboveground vascular biomass of individual species yielded some similar results (Fig. 4). The increase in live *E. scheuchzeri* was significant at $p = 0.0005$, but in the same plot the concomitant reductions in *Carex* and *E. angustifolium* were not significant ($p = 0.10$). However, the dead standing crop of *Carex* was significantly reduced on both experimental plots ($p = 0.025$). Live *Carex* was significantly reduced ($p = 0.005$) on the experimental plot with standing water where no bryophytes were added. On an adjacent, naturally recovering track with standing water (Plot 3), only the live biomass of *E. angustifolium* was reduced ($p = 0.025$), but overall species richness was reduced by half compared to its control (Forbes 1992a). Species richness in Treatment 1 was nearly equal to its control (6 versus 7), whereas there were only three species present in Treatment 2 with standing water, compared to seven in the control.

Among nonvascular plants, the multivariate analyses detected patchy qualitative differences in the frequencies of bryophytes present between treatments and controls. These differences were apparent among both naturally recovering tracks and the experimental

plots, but no clear patterns within species emerged, so that differences were attributed to random processes of local dispersal in the case of the plot where bryophytes were added. The only important environmental difference detected between treatments and controls was an increase in mean organic mat depth on both experimental plots. However, this increase (≈ 4 cm) was significant only on the plot where bryophytes were added ($p = 0.025$). There were no differences in pH or corrected conductivity between disturbed plots and controls.

Discussion

The population of *Eriophorum scheuchzeri* occurring in the plot with flowing water was unusual in two respects. For one, the plants were in full flower, while few of the countless culms of *Eriophorum angustifolium* from nearby controls bore even marginally developed inflorescences, as did few of the equally countless shoots of *Carex aquatilis* var. *stans*. For another, the shape of the densely-flowering patch conformed almost precisely to the dimensions of the restored sample plot. The population of *E. scheuchzeri* seems clearly to have germinated from seeds that were present in the seed bank at the time of transplanting. This conclusion is supported by both the distribution of the present population and by the fact that there were no individuals occurring on the adjacent experimental plots comprised of vegetative shoots only. How old the viable seeds were at the time of germination is unknown. Reports of great seed longevity in the Arctic have been suspect (Porsild et al. 1967), but it is notable

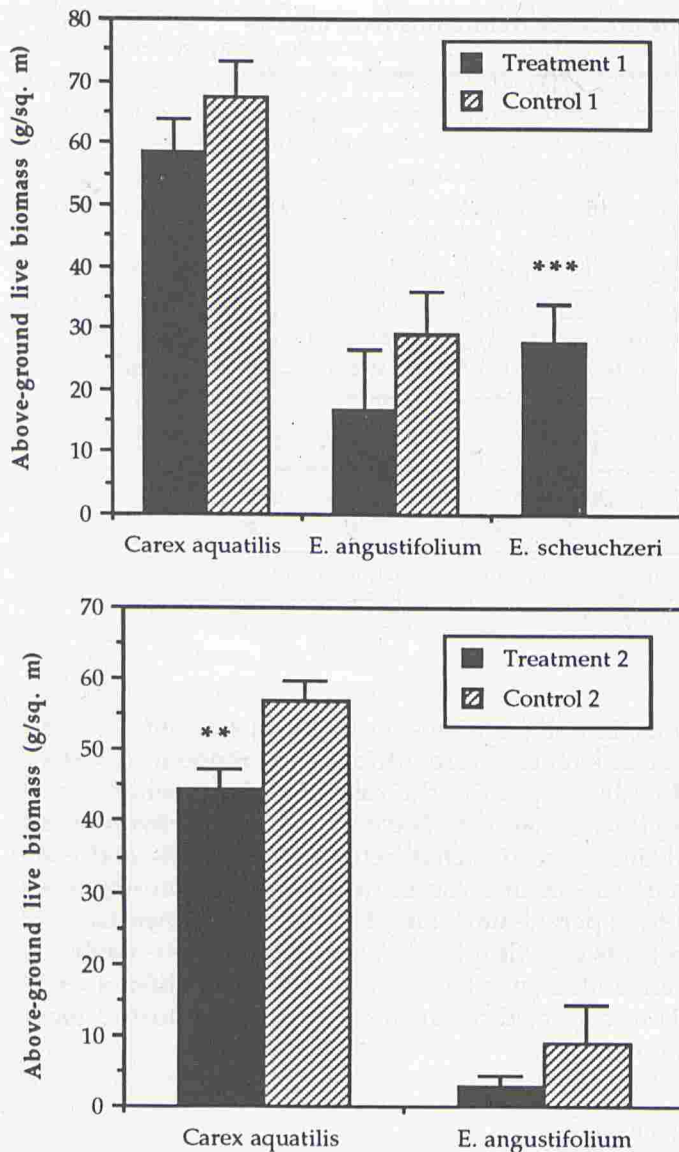


Figure 4. Comparison of sedge biomass ($\pm$ S.E.) in treatments and adjacent controls 18 years after initial transplants. Histogram at top depicts plot (Treatment 1) with flowing water, in which both *Carex* shoots and bryophyte sods were transplanted, and adjacent control. Bottom histogram depicts plot with standing water (Treatment 2) in which only *Carex* shoots were transplanted, and adjacent control. Species are *Carex aquatilis* var. *stans*, *Eriophorum angustifolium*, and *E. scheuchzeri*. *** and ** indicate significant difference from control at $p = 0.0005$ and 0.005 respectively. Measurements made July 28, 1990.

that Alaskan seeds of *Carex bigelowii* (Bigelow's sedge), reliably dated to be about 200 years old (± 80 years), were recently germinated by McGraw et al. (1991). Although little is yet known about the dynamics of tundra seed banks, it seems that more than half of all tundra species regularly produce seed that is retained

in the soil. These seed banks are assumed to have low turnover rates, with seeds that have correspondingly long persistence times (McGraw & Vavrek 1989).

There are apparently few data on the storage of seeds in the soil, viable or otherwise from the high arctic (McGraw & Vavrek 1989). However, counts made on naturally recovering disturbances here and on Baffin Island yielded an almost insignificant number of seedlings, most of which were either *Salix arctica* (arctic willow) or colonizing grasses, such as *Phippsia algida* (snow goosegrass) and *Alopecurus alpinus* (alpine fox-tail) (Forbes 1992a). The latter two species were absent from the natural vegetation of the Truelove Lowland. In contrast, studies of similar disturbances in the Low Arctic show that recruitment from buried seed can be by far the major source of colonizers on a given patch (Hernandez 1973; Ebersole 1989; J. B. McGraw 1990, personal communication).

Germination among the sedges present, other than *E. scheuchzeri*, is uncommon within the region, and no seedlings of these species were observed. Bliss and Grulke (1988) have noted that, for high arctic populations of *C. aquatilis* var. *stans*, from 6% to 9% of the shoots may flower per year but few if any of the seeds are viable. At the time of the initial manipulation, Barrett (1975) counted no mature seeds in 125 pistillate inflorescences on 100 flowering stems of this species in a similar meadow on Truelove Lowland. Seedlings of *E. angustifolium* are also quite rare and, even in the temperate zone, bare areas are invariably colonized by rhizomes (Phillips 1954).

The plot with standing water and with only vegetative *Carex* shoots (no moss) used, also looked entirely restored at a glance and blended in well with the surrounding matrix of undisturbed vegetation. Growth of both *Carex* and *E. angustifolium* appeared vigorous, but inflorescences were poorly developed compared to controls. As the above results made clear, however, neither vascular cover/abundance nor biomass had been restored to predisturbance levels. It is interesting to note that in the flowing water plot where *E. scheuchzeri* had become established (Treatment 1, Fig. 5), live leaf cover of both of the other sedges was significantly reduced but their live biomass was not, compared to the controls (Fig. 4). This indicates that the weight of individual shoots of *Carex* and *E. angustifolium* was higher on the experimental plots than on the control plots.

It is notable that total plant cover in the control plots with standing water (Treatment 2 and Plot 3) was only half that measured in the plot with seasonally flowing water (Table 1). Although it is not certain, these differences may be at least partly related to spatial and temporal variation in available nutrients provided from surface runoff, as documented in other high arctic wet-



Figure 5. Experimental plot (Treatment 1) in which both vegetative shoots of *Carex aquatilis* var. *stans* and bryophyte sods were applied (Fig. 2) to severely damaged ruts. Inflorescences of *Eriophorum scheuchzeri*, a species completely lacking in the undisturbed plots, are conspicuous. Note the reduction in cover of both live and dead standing leaf tissue of *Carex* and *E. angustifolium* compared to Figure 6. Quadrat = 25 × 50 cm.

lands (Murray & Svoboda 1989; Buttle & Fraser 1992). It is relevant that Chapin et al. (1988) also found seasonally flowing water to enhance productivity and nutrient cycling at highly local scales in low arctic tundra.

In the naturally recovering rut with standing water, levels of vascular cover/abundance and biomass have failed to fully recover in approximately 19 years, most notably in *E. angustifolium* (Forbes 1992a). Although total plant cover was lower in all the disturbed plots than in the controls, the proportional reduction was least in the two restored plots (Table 1). Thus, in terms of restoring a binding vascular cover, the restoration effort was an improvement over natural recovery of vehicle tracks on the lowland (Forbes 1992a). That the bryophyte composition of all three disturbed plots differed slightly from the controls was not considered significant because there was not consistent variation detectable and because bryomass appeared more than adequate to restore the hydrological status of the disturbed patches. Water flow rates, although not measured, were visibly reduced in the sodded plot relative to adjacent untreated portions of the same set of tracks.

One of the more interesting results is that the moss-sodded plot is quite different in several ways from the other treatment and from the naturally recovering ruts, including the population of *E. scheuchzeri* apparently recruited from seed. Controlled studies with Alaskan populations of this plant have found it to be adapted to more fertile soils than *Eriophorum vaginatum* (hare's tail grass) and its flowering responses linked to nutrient availability (McGraw & Chapin 1989). Thus, the persistence and apparent vigor of the lone popula-

tion of *E. scheuchzeri* on the restored plot after 17 years may be a signal that the local nutrient regime is more favorable on the experimental plot than in the adjacent undisturbed stands. This conclusion may be reinforced by the small but significant increase in cover of *Ranunculus hyperboreus* (Table 1), considered to be a nitrophile, on this plot (Porsild & Cody 1980).

The unexpected germination of the population of *E. scheuchzeri* provides yet another example of the importance of initial floristics, as discussed in the works of Egler (1954) and Kenfield (1991). Regardless of when the seeds of *E. scheuchzeri* actually germinated, this would appear to be an example of complete initial floristics, in which there are no further species arrivals after the establishment of the sedge (Wilson et al. 1992). Others have also shown, from studies in a variety of ecosystems, that it is often the case that vegetation change on a given patch will be directed largely by those species present from the outset at the cessation of disturbance, rather than by subsequent colonizers (Jha & Singh 1991; Melick & Ashton 1991). These results highlight the potential significance of the seed bank, at least for arctic wetlands where assisted recovery involves substrate transplants, as in the case of the bryophyte sods. The findings are thus encouraging in that they give an indication of where we need to focus our efforts. The seed bank will likely prove to be the single most important source of unexpected responses in any efforts that involve disruption of even highly local substrates, as was the case in this study. I hereby suggest the need for a regional seed atlas. This effort could be coupled with ongoing mapping of the bioclimatic zones (Edlund 1990), and studies such as

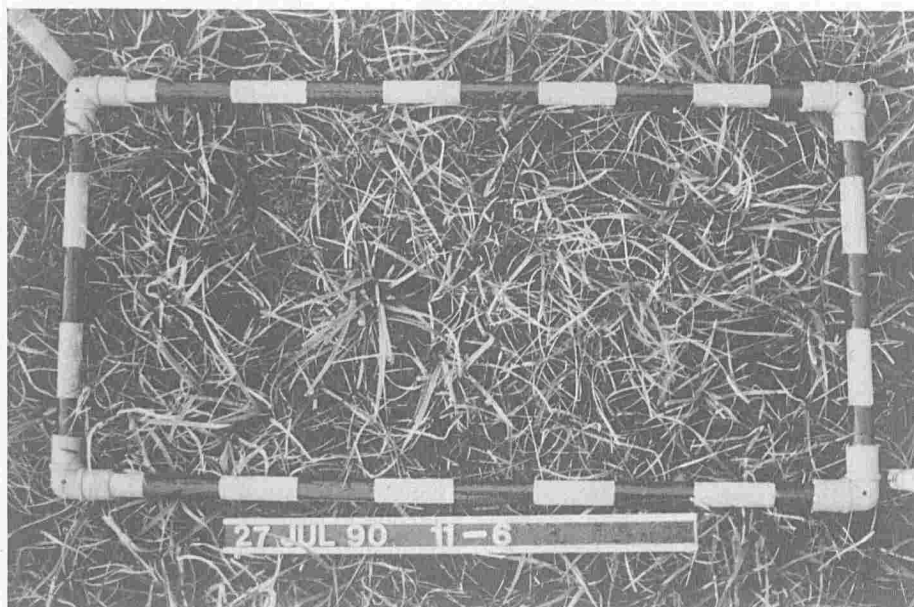


Figure 6. Control plot in undisturbed portion of wet meadow dominated by *Carex aquatilis* var. *stans* with lesser amounts of *Eriophorum angustifolium*. Note the abundance of dead leaf tissue, seen here as lighter, thread-like material, and the lack of inflorescences. Quadrat = 25 × 50 cm.

those by Mitchell and McKendrick (1975) in which the general climatic requirements are determined for where and when common species bear viable seed. In truth, enough work has already been done to make it abundantly clear that summer warmth is considered the dominant macroenvironmental control delimiting gross distributions of vascular plants in the Arctic (Young 1971; Edlund 1990), and that a mosaic rather than a zonal pattern is most appropriate to understanding intra- or subregional biological diversity (Matveyeva 1988).

Has this restoration effort been a success? It is important to consider scale effects in assessing this and other restoration efforts. Meadow hydrology and function appear fully restored in planted plots. Within the disturbed plots, species richness approaches that of controls only in the restored rut with flowing water (Table 1). It is not clear if this difference is correlated with the flowing water, the addition of bryophyte sods, or some other factor. Species composition appears indefinitely, if unintentionally, altered in the case of the dense and profusely flowering population of *E. scheuchzeri*. In effect, a single unexpected sedge recruited from the seed bank has persisted on the site for perhaps 15 years and has competed for space but not adversely reduced the biomass of the other sedges. In separate mesic sites on the lowland, disturbed in 1986, many ruderal species germinated within one year and established a dense cover within three years, indicating that stores of viable seed exist and may give rise to plants that can quickly colonize anthropogenic patches.

In the long term, the successful restoration of a bind-

ing plant cover in the ruts may reflect the abilities for vegetative regeneration among the various vascular (Bliss & Grulke 1988) and nonvascular (Longton 1988) components of the limited flora. Proportional reductions in total plant cover were less apparent in restored ruts than in naturally recovering ruts. Given this perspective, how much do the highly local but significant reductions in biomass and cover detected among dominant vascular plants matter? This depends on the scale of the initial disturbance and the scale and aims of the attempted restoration effort. As it is, these and many other lowlands in the region comprise critical grazing habitat for muskox and other herbivores. At present, however, the extent of impacts such as these fortunately remains quite limited. Also, the general lack of disturbance on slopes of $\geq 2^\circ$ has meant that drainage, erosion, and other hydrological effects have not been extensive. At the regional scale, anthropogenic tundra disturbance may increase in the future, and experiments such as these will prove crucial to our understanding of what manipulation can accomplish under the current macroclimatic regime.

In the end, a battery of measurements and statistical tests detected few biotic or abiotic differences between restored plots and adjacent controls. However, the establishment, persistence, and competitive success of a discrete population of *E. scheuchzeri*, derived from seed present in restorative bryophyte sods, is important. The germination of this species serves as a small but crucial reminder of the potential for the unexpected to arise when dealing with manipulations involving either intentional exploitation or unintentional disruption of seed banks. It would be prudent to expect that

stores of viable seed will be somewhat greater in arctic wetlands at lower latitudes, where a greater proportion of the dominant species regularly set viable seed.

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